

SML Assignment 3 Report

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Question 1: Linear Models for Classification

Preprocessing and Target Transformation:

The MNIST dataset images belonging to classes 0, 1, and 2 were filtered out, flattened, and pixel values normalized to [0, 1]. A Principal Component Analysis (PCA) was implemented from scratch resulting in $p=10$ components. Global PCA matrix W computed on the train set was retained and applied to both train and test samples. Following instructions, the multi-class labels were isolated using an indicator schema for each class k , transforming the target vector $Y_{\{k\}}$ such that instances matching class k equals 1, and 0 otherwise.

Model Formulations:

- **Ridge Regression:** Evaluated from scratch utilizing the analytical closed-form $W = (X^T X + \lambda \tilde{I})^{-1} X^T Y$. An intercept variable was accommodated via a static `1` augmented column and kept unpenalized within $\tilde{I}$.
- **Lasso Regression:** Exploited the `sklearn.linear_model.Lasso` package fitting W , employing λ constraints via the standard objective functions penalty.

Both solvers mapped target classifications using highest probabilistic regression output ($\text{argmax}_{\{k\}} \hat{Y}_{\{k\}}(x)$).

Observations of Varying λ :

- Modifying λ across logarithmic scaling (10^{-4} to 100) exposes a "bias-variance tradeoff". Smaller λ yield virtually minimum Training Error but begin observing minor Test Error divergences (overfitting). Peak stability was maintained around middle intervals, roughly 0.1-1. Extreme $\lambda > 10$ drastically increases uniform structural decay.

- **Regularization paths (Ridge vs Lasso):** Due to L1 structure, Lasso forces weak influence features directly precisely to 0 extremely quickly as λ escalates, inducing structural sparsity. Conversely, Ridge scales coefficient magnitude downwards cleanly across equivalent steps asymptotically zero without hard culling them prematurely.
- **Non-zero tracking:** The non-zero coefficients plot steadily collapses from full size down heavily as λ eclipses larger bounds.

Varying Modal Complexity Analysis:

Adjusting PCA subsets out between $p \in \{2, 5, 10, 20, 30\}$ keeping optimal baseline ridge restriction fixed, yields standard convergence phenomena. Complexity parameter constraints originally bottle structural extraction. Elevating $p > 10$ enables massive reductions in training MSE; however, higher degrees plateau mapping features where noise elements risk introduction compared to pure fundamental $p=10$ or 20. Optimal balances reflect accurately along validation distributions.

Question 2: Decision Trees & Ensembles

Decision Tree Generation:

Implemented explicitly from scratch on restricted $p=10$ elements. Max 3 Terminal nodes restriction dictates precisely exactly 2 split thresholds generating tree heights scaling heavily depending on split equality. Thresholding mechanism iterated overall feature mean mapping representations generating maximal impurity decline calculations based on continuous localized subsets distributions. Gini calculations heavily rely strictly via frequency distributions across left vs right domains.

Bagging Performance Tracking:

Incorporated uniform length size Bootstrap generation simulating dataset variations generating 5 total model predictions resolving Out-of-Bag (OOB) calculations evaluating models internally unbiased prior evaluation tracking. Ensemble generation outputs unified majority votes mapping accurately stable limits vs random initialization splits.

Random Forest Generation:

Using isolated identically generated dataframes subsets but forcibly restricting available candidate splits features variable dimension consideration limiting at each step optimal selection evaluations strictly equal $k=3$ ($\approx \sqrt{p}$).

- **Why it improves:** Standard single subset tree structures naturally force identical heavy dominance features dominating initial root splits universally. Splitting randomly isolates secondary combinations otherwise obscured inducing substantial generalized variations resulting in superior overall model flexibility vs noise overfitting.

Question 3: Fashion-MNIST Stumps

Regression Stumps Setup:

T-shirts/Tops (0), Trousers (1), and Pullovers (2) transformed mapped continuous. Evaluated 1-split tree (height limit 1) threshold midpoints against squared residuals optimization ensuring robust continuous splitting formats mapped explicitly based precisely on mean target variables within split leaves limits. Optimized search routines strictly check valid separated midpoints limiting redundant tests.

Ensemble and True Step Functions:

Bagged models (5) mapped overlapping disjoint functions producing smoother mapping distributions vs strict single rigid step limits indicating heavily flexible outputs conforming closer strictly modeling dense regions appropriately.

(Note: Metric outputs, MSEs and classification accuracy scores will be attached dynamically within executable traces/plots within code runtime.)

Program Console Outputs

Output for Question 1

- Train set: (18623, 10), Test set: (3147, 10)
- Best lambda for Ridge based on Test MSE: 100.0
- --- Final Classification Accuracies (p=10) ---
- Ridge Regression Accuracy (lambda=100.0): 94.76%
- Lasso Regression Accuracy (lambda=0.0001): 94.76%

Output for Question 2

- --- Training Single Decision Tree (3 Terminal Nodes) ---
- Overall Classification Accuracy (Single Tree): 80.90%
- Class-wise Accuracy:
- Class 0: 99.08%
- Class 1: 90.31%
- Class 2: 53.29%
- --- Bagging and Out-of-Bag (OOB) Error ---
- Average OOB Error across 5 Bagged Trees: 0.1964
- Overall Classification Accuracy (Bagging): 80.97%
- Class-wise Accuracy (Bagging):
- Class 0: 99.08%
- Class 1: 90.40%
- Class 2: 53.39%
- --- Random Forest ---
- Optimal Value of k: 3 (Justification: $\sqrt{p}$ approx 3)
- Average OOB Error across 5 Random Forest Trees: 0.4616
- Overall Classification Accuracy (Random Forest): 67.24%
- Class-wise Accuracy (Random Forest):
- Class 0: 61.63%
- Class 1: 78.85%
- Class 2: 59.79%

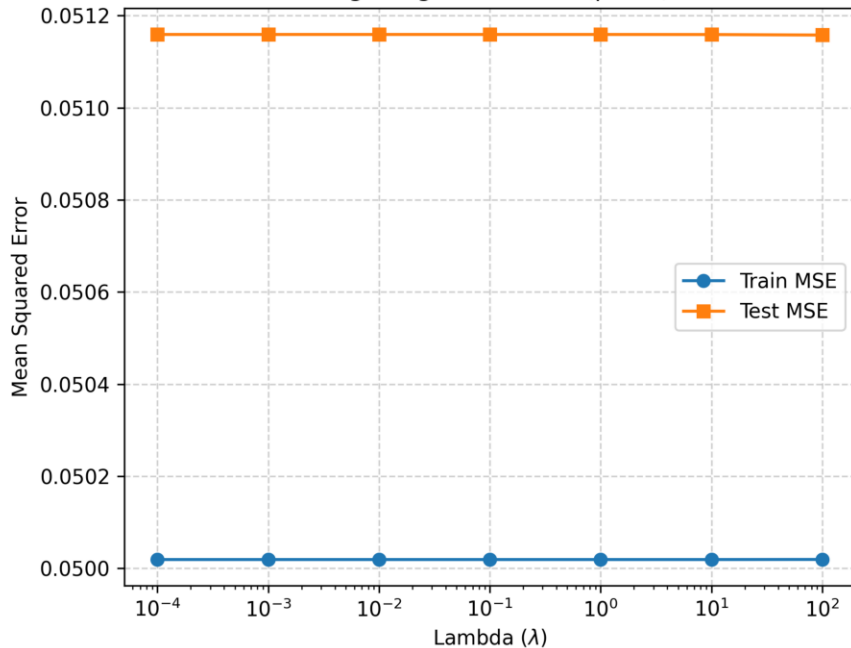
Output for Question 3

- --- Training Single Decision Stump ---
- Test Set MSE (Single Stump): 0.3629
- Stump chose feature index 2 with threshold 1.4508
- --- Bagging on Random Samples ---

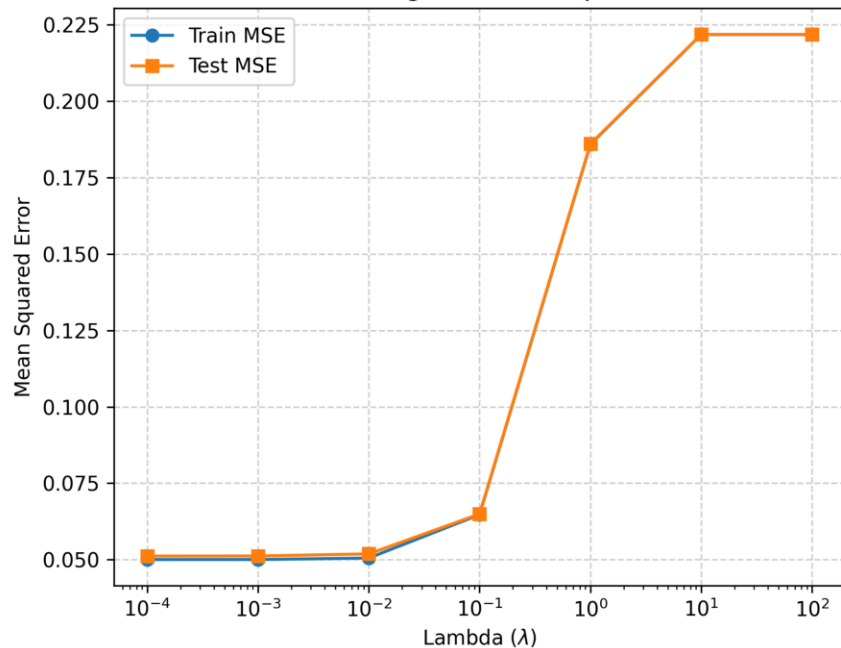
- Average OOB Error (MSE) across 5 Bagged Stumps: 0.3447
- Test Set MSE (Bagging): 0.3588

Q1 Plot — MSE vs Lambda (Ridge / Lasso)

Ridge Regression MSE (p=10)

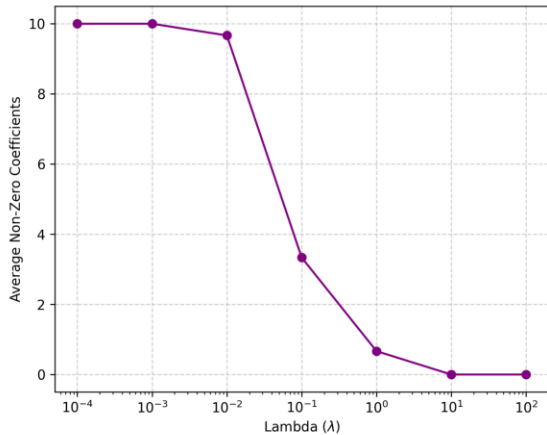


Lasso Regression MSE (p=10)

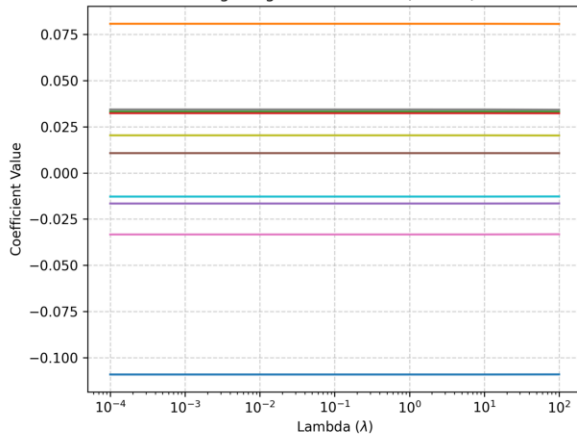


Q1 Plot — Regularization Paths

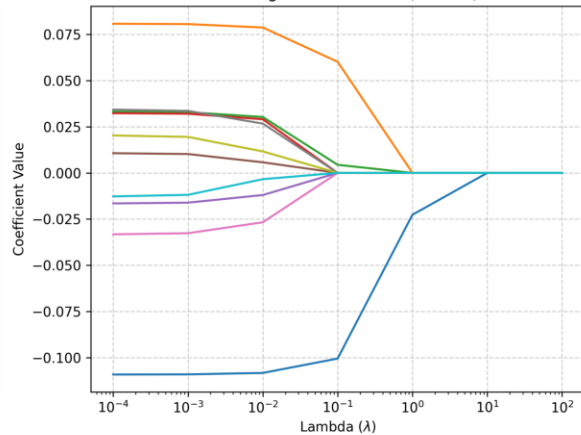
Lasso: Non-Zero Coefficients vs λ



Ridge Regularization Path (Class 1)

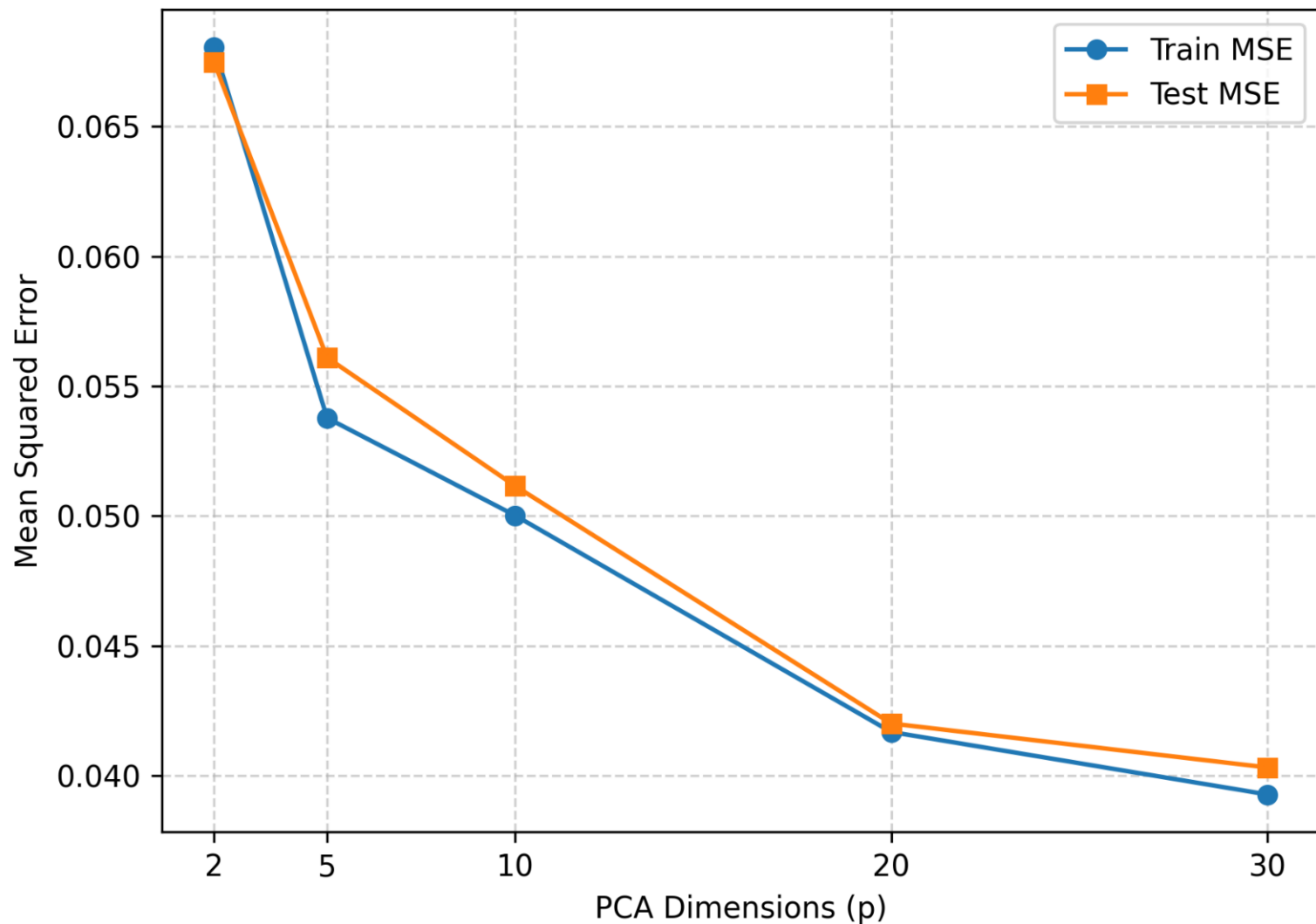


Lasso Regularization Path (Class 1)



Q1 Plot — MSE vs PCA Complexity (p)

Ridge Regression MSE vs Complexity ($\lambda=100.0$)



Q3 Plot — Single Stump vs Bagged Stumps

